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CODE:

```
# Naive Bayes Classification for Bank Loan Prediction

import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report
)

# Sample Bank Loan Dataset
data = {
    "Age": [
        25, 30, 35, 40, 45,
        50, 28, 32, 38, 42,
        48, 55, 27, 36, 52
    ],

    "Income": [
        25000, 40000, 55000, 60000, 75000,
        90000, 30000, 45000, 50000, 65000,
        80000, 100000, 28000, 52000, 85000
    ],

    "Credit_Score": [
        600, 650, 700, 720, 750,
        800, 620, 680, 690, 730,
        770, 820, 580, 710, 780
    ],

    "Loan_Amount": [
        100000, 150000, 200000, 250000, 300000,
        350000, 120000, 180000, 220000, 270000,
        320000, 400000, 90000, 210000, 330000
    ],

    "Loan_Approved": [
        0, 0, 1, 1, 1,
        1, 0, 1, 1, 1,
        1, 1, 0, 1, 1
    ]
}

# Convert dataset to DataFrame
df = pd.DataFrame(data)

print("BANK LOAN DATASET")
print("=" * 50)
print(df)

# Input features
X = df[[
    "Age",
    "Income",
    "Credit_Score",
```

```
    "Loan_Amount"
]]

# Target variable
y = df["Loan_Approved"]

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Create Gaussian Naive Bayes model
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display results
print("\nNAIVE BAYES BANK LOAN PREDICTION RESULTS")
print("=" * 50)

print("\nAccuracy:", round(accuracy * 100, 2), "%")

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=["Not Approved", "Approved"],
    zero_division=0
))

# Predict loan approval for a new customer
new_customer = pd.DataFrame({
    "Age": [35],
    "Income": [65000],
    "Credit_Score": [720],
    "Loan_Amount": [200000]
})

prediction = model.predict(new_customer)

print("\nNEW CUSTOMER DETAILS")
print("=" * 50)

print("Age:", new_customer["Age"].iloc[0])
print("Income: ₹", new_customer["Income"].iloc[0])
print("Credit Score:", new_customer["Credit_Score"].iloc[0])
print("Loan Amount: ₹", new_customer["Loan_Amount"].iloc[0])

if prediction[0] == 1:
    print("\nLoan Prediction: APPROVED")
```

```
else:
    print("\nLoan Prediction: NOT APPROVED")
```

OUTPUT :

```
BANK LOAN DATASET
==================================================
    Age  Income  Credit_Score  Loan_Amount  Loan_Approved
0    25   25000           600       100000              0
1    30   40000           650       150000              0
2    35   55000           700       200000              1
3    40   60000           720       250000              1
4    45   75000           750       300000              1
5    50   90000           800       350000              1
6    28   30000           620       120000              0
7    32   45000           680       180000              1
8    38   50000           690       220000              1
9    42   65000           730       270000              1
10   48   80000           770       320000              1
11   55  100000           820       400000              1
12   27   28000           580        90000              0
13   36   52000           710       210000              1
14   52   85000           780       330000              1

NAIVE BAYES BANK LOAN PREDICTION RESULTS
==================================================

Accuracy: 100.0 %

Confusion Matrix:
[[1 0]
 [0 2]]

Classification Report:
              precision    recall  f1-score   support

Not Approved       1.00      1.00      1.00         1
    Approved       1.00      1.00      1.00         2

    accuracy                           1.00         3
   macro avg       1.00      1.00      1.00         3
weighted avg       1.00      1.00      1.00         3


NEW CUSTOMER DETAILS
==================================================
Age: 35
Income: ₹ 65000
Credit Score: 720
Loan Amount: ₹ 200000
```